

Text Classification: From TF-IDF to PyTorch

A controlled comparison of symbolic, dense and neural text representations

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This project evaluates how document representation changes classification performance on the 20 Newsgroups benchmark. The same six-category task is tested with Bag of Words, TF-IDF, topic-model reductions, count-based and prediction-based embeddings, and PyTorch averaging networks. Stored notebook outputs show that fine-tuned GloVe embeddings produced the strongest held-out result: **80.12% test accuracy**.

80.12%

BEST HELD-OUT ACCURACY

7,309

DOCUMENTS IN THE TEST SET

6

HIGH-LEVEL CLASSES

Method and implementation

- Removed headers, footers and quoted replies before modelling to reduce author- and thread-specific leakage.
- Used a common 8,000-token vocabulary and separate training, validation and held-out test partitions.
- Compared Bag of Words and TF-IDF with logistic regression; SVD/LSA and LDA topic reductions; PPMI, Word2Vec and 300-dimensional GloVe embeddings.
- Implemented custom PyTorch Dataset/DataLoader components and averaging classifiers, including frozen and fine-tuned pretrained embeddings.
- Reviewed class-level confusion patterns and used validation loss for early stopping rather than selecting directly on the test set.

Selected stored results

Representation / model	Held-out test accuracy
Bag of Words + logistic regression	75.11%
TF-IDF + logistic regression	78.77%
SVD / LSA + logistic regression	76.47%
LDA + logistic regression	68.08%
PyTorch averaging model + fine-tuned GloVe	80.12%

Values above are transcribed from outputs stored in the executed notebook; they were not recomputed for this portfolio brief.

Interpretation

TF-IDF remained a strong and efficient baseline. Fine-tuning pretrained embeddings improved the neural model enough to exceed that baseline by 1.35 percentage points on the held-out set. The comparison also shows that dimensionality reduction does not automatically improve the task: LDA lost substantial discriminative information, while SVD remained closer to the sparse baseline.

Scope and limitations

- This is an individual academic laboratory on a standard benchmark, not a production deployment or a newly collected dataset.
- No statistical significance test was stored, so small differences between the strongest models should be interpreted cautiously.
- The original notebook contains raw sample posts from the public dataset. It is intentionally not attached; this sanitized brief excludes emails, phone numbers, addresses and long raw-text outputs.

Evidence source: *Lab_Classification_4AI12_Manuel_David_MAYA_ROSERO.ipynb*. This text-only brief was prepared for portfolio publication from the notebook's saved code, analysis and outputs.